Robust Open-Set CAN Attack Classification under Domain Shift: A Dual-Alignment Framework

Wentao Li[1], Jianfeng Yang[1], Jianling Hu[1,2] and Tianqi Yu[1]*
1. School of Electronic and Information Engineering, Soochow University, Suzhou, China
2. School of Electronic and Information Engineering, Wuxi University, Wuxi, China
*Corresponding Author: tqyu@suda.edu.cn

***Abstract*—To address domain shift in in-vehicle CAN and the challenge of identifying unknown attacks in open-set scenarios, this paper proposes a cross-domain intrusion detection network (DORU-Net) based on dual alignment and open-set rejection. First, CAN messages are converted to grayscale images for convolutional neural network-based high-dimensional spatiotemporal feature extraction. A DANN-based architecture integrated with GRL and CORAL loss is then constructed for feature alignment, realizing dual constraints on source and target domains in both second-order statistical distribution and adversarial feature space to boost alignment robustness. In inference, a cascaded "known class determination—OpenMax adaptive rejection" mechanism is designed, which uniformly identifies single or mixed unknown attacks in the target domain as "unknown classes" via distribution boundaries. DORU-Net thus forms a robust security monitoring framework for Non-IID data in connected vehicle systems.**

***Index Terms*—CAN, Intrusion Detection, Domain Adaptation, Open-Set Recognition**

I. Introduction

With the advancement of Intelligent Connected Vehicles and autonomous driving technologies, each vehicle has evolved as a cyber-physical node, and the entire vehicular ecosystem has further formed a large-scale interconnected Cyber-Physical System. As the backbone of in-vehicle communication, the Controller Area Network (CAN) is responsible for transmitting critical commands such as powertrain, chassis, and body control [1]. However, the CAN protocol lacked encryption and authentication mechanisms at its inception, making it highly vulnerable to malicious attacks such as spoofed message injection and replay, which pose serious threats to vehicle security [2].

Although deep learning-based Intrusion Detection Systems (IDS) have achieved remarkable performance in attack identification for dedicated vehicle models [3], their cross-vehicle generalization is severely constrained by domain heterogeneity. For CAN intrusion detection, **cross-domain** characterizes the distinct data distributions derived from disparate vehicle models, signal coding specifications, CAN ID allocation schemes and driving scenarios. The **source domain** is defined as the labeled CAN dataset for model training, and the **target domain** as the unseen vehicle platforms and practical driving environments where the detection model is deployed. Against this background, two critical bottlenecks remain unresolved in real-world applications:

1) Domain Shift caused by Non-IID Data: CAN messages under different vehicle models or operating conditions exhibit significant differences in signal encoding and ID distribution. When models trained in the source domain are directly transferred to the target domain, performance often degrades drastically due to mismatched feature distributions.
2) Open-Set Recognition Problem: Existing methods mostly rely on the closed-set assumption, where training and testing sets share the same label space [4]. However, in real-world scenarios, the target domain often encounters unknown attacks never seen in the source domain. Traditional classifiers tend to force these unknown samples into known categories, leading to high-risk misclassifications.

Addressing the aforementioned challenges, this paper proposes a Domain-Adaptive Open-Set Rejection-based Unified Cross-Domain Intrusion Detection Network (DORU-Net). This framework aims to achieve precise migration of known attacks while effectively intercepting unknown threats without requiring labels from the target domain. The main contributions of this paper are as follows:

1) **Construction of a Dual Feature Alignment Architecture:** An adversarial adaptation network integrating a Gradient Reversal Layer (GRL) with CORAL loss is proposed [6]. The GRL achieves first-order distribution alignment of the feature space, while CORAL constrains second-order statistics. Their synergistic operation significantly enhances the feature invariance and robustness of the model in cross-vehicle scenarios.
2) **Design of a Cascaded Open-Set Inference Mechanism:** An innovative inference strategy is designed, comprising "Known Class Determination" followed by "OpenMax Adaptive Rejection." Utilizing Extreme Value Theory (EVT) to fit the tail distribution of activation vectors, The probability of a sample belonging to an unknown class is calculated [5]. This mechanism ensures recognition accuracy for known attacks while flexibly intercepting unseen attacks as generic unknown threats.

The rest of the paper is organized as follows. Section II elaborates on the preliminary work. Section III details the proposed model architecture. Section IV presents and discusses the evaluation of the experimental results. Finally, Section V summarizes the whole paper and proposes future research directions.

II. Preliminaries

This section presents the fundamental theories and key technologies underpinning DORU-Net, focusing on four core aspects to support subsequent model design and experiments. It first elaborates on the CAN protocol, its message structure, common attack types, and utilized datasets, providing a practical background for in-vehicle network intrusion detection. Then, it explains domain adaptation, the DANN framework, and the role of GRL in addressing cross-vehicle domain shift. Next, the CORAL method is introduced to complement feature distribution statistical alignment. Finally, it covers open-set recognition and the OpenMax algorithm, laying the theoretical foundation for the model to handle unseen threats.

A. Controller Area Network (CAN)

1) Message Structure: Controller Area Network is a broadcast-oriented, multi-master bus communication protocol widely used for data exchange between automotive electronic control units (ECUs). The key fields of a standard CAN message are shown in Table I.

TABLE I: Frame Structure of CAN Message

Start of Frame	Arbitration Field	Control Field	Data Field	CRC	ACK	End of Frame
1 Bit	12 Bits	6 Bits	0-64 Bits	16 Bits	2 Bits	7 Bits

2) CAN Attack Types and Threat Model: The specific characteristics and potential consequences of these In-Vehicle Network (IVN) attacks are detailed in the TABLE II.

TABLE II: Description of Attack Types and Normal Traffic

Traffic Type	Description
DoS Attack	Floods the network with high-priority messages to block normal traffic transmission, causing potential loss of control.
Fuzzy Attack	Injects messages with arbitrary IDs and data (sometimes mimicking valid IDs) to disrupt system stability unpredictably.
Gear Spoofing	Floods the bus with gear-control related IDs and random data to mislead the ECU regarding gear operations.
RPM Spoofing	Overrides legitimate signals using tachometer-linked IDs to manipulate displayed RPM readings on the dashboard.
Malfunction Attack	Modifies data fields of specific extracted IDs and injects them to trigger abnormal vehicle behaviors.
Normal Traffic	Standard network traffic generated during the vehicle's regular operation without malicious interference.

In this work, it assumes that the source domain only includes known categories such as normal packets and DoS attacks, while the target domain additionally includes attack types not seen in the source domain, such as fuzzy attacks and malfunction attacks, thus forming an open-set domain adaptation scenario.

B. Domain Adaptation

1) Definition of the domain adaptation problem: Let the source domain be defined as

$$D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s} \tag{1}$$

where D_s denotes the labeled source domain dataset, where each element (x_i^s, y_i^s) corresponds to one sample from the source domain: x_i^s is the feature vector of the i-th source sample, y_i^s is its category label, and n_s is the total number of samples in D_s. All these samples are independently and identically drawn from the joint probability distribution $P_s(X, Y)$.

The target domain dataset is defined as

$$D_t = \{x_j^t\}_{j=1}^{n_t} \tag{2}$$

where D_t denotes the unlabeled target domain dataset, where each x_j^t is the feature vector of the j-th target sample, and n_t is the total number of samples in D_t. These samples are independently and identically drawn from the joint probability distribution $P_t(X, Y)$, and their category labels are typically unavailable or only sparsely provided.

Due to the discrepancies in CAN frame ID distributions and data characteristics across distinct vehicle models, this commonly leads to

$$P_s(X, Y) \neq P_t(X, Y) \tag{3}$$

which constitutes the domain shift problem.

Unsupervised domain adaptation (UDA) aims to boost target-domain classification accuracy by learning a feature extractor $F(\cdot)$ and a classifier $C(\cdot)$ with labeled source data and unlabeled target data. A core principle is to align cross-domain feature distributions via $F(\cdot)$, thereby improving the generalization of $C(\cdot)$ on the target domain [**?**].

2) Domain-Adversarial Neural Network: DANN is a typical adversarial domain adaptation framework. The core idea is to simultaneously train two modules on a shared feature extractor F.

1. Source Domain Classifier C performs supervised classification on features from the source domain.

2. Domain Discriminator G determines whether the input features originate from the source domain or the target domain.

Through adversarial training, the objective is for the feature extractor F to learn features that are both effective for source domain task classification and difficult for the domain discriminator G to determine their origin. This enables the learning of domain-invariant representations.

In the DANN framework, the source domain classification loss is defined as

$$L_{\text{cls}} = \frac{1}{n_s} \sum_{i=1}^{n_s} l(C(F(x_i^s)), y_i^s) \tag{4}$$

where $l(\cdot)$ is typically the cross-entropy loss.

The domain discrimination loss is defined as

$$L_{\text{dom}} = \frac{1}{n_s + n_t} \sum_{k} l(G(F(x_k)), g_k) \tag{5}$$

where $g_k \in \{0, 1\}$ is the domain label, where $g_k = 0$ denotes the source domain and $g_k = 1$ denotes the target domain.

3) Gradient Reversal Layer: To achieve the adversarial training described above, DANN introduces a Gradient Reversal Layer (GRL) [6]. During forward propagation, GRL acts as an identity mapping

$$\mathrm{GRL}(z) = z \tag{6}$$

During backward propagation, however, GRL multiplies the gradient passing through it by a negative coefficient $-\lambda$

$$\frac{\partial \mathrm{GRL}}{\partial z} = -\lambda I \tag{7}$$

where I is the identity matrix.

During joint training, the domain discriminator D minimizes the domain discrimination loss L_{dom}, while the feature extractor F, influenced by the GRL, effectively maximizes L_{dom}, pushing the source and target feature distributions to become indistinguishable.

The overall optimization objective can be expressed as

$$\min_{F,C} \max_{D} \quad L_{\text{cls}}(F, C) - \lambda L_{\text{dom}}(F, D) \tag{8}$$

Building upon this framework, the statistical alignment loss CORAL is further introduced to construct a dual-alignment training strategy.

C. Correlation Alignment

Correlation Alignment (CORAL) [7] is a method for transfer learning that explicitly matches the second-order statistics of features between the source domain and the target domain. The core idea is to align not only the means of feature distributions in the feature space but also their covariance matrices.

Let the feature matrix of the source domain be $H_s \in \mathbb{R}^{n_s \times m}$, and the feature matrix of the target domain be $H_t \in \mathbb{R}^{n_t \times m}$, where m is the feature dimension. The covariance matrices of the source and target domains can be calculated as

$$\Sigma_s = \frac{1}{n_s - 1}\left(H_s^\top H_s - \frac{1}{n_s}(1^\top H_s)^\top (1^\top H_s)\right) \tag{9}$$

$$\Sigma_t = \frac{1}{n_t - 1}\left(H_t^\top H_t - \frac{1}{n_t}(1^\top H_t)^\top (1^\top H_t)\right) \tag{10}$$

where 1 represents a vector of all ones.

The CORAL loss is defined as the Frobenius norm distance between the two covariance matrices

$$L_{\text{coral}} = \frac{1}{4m^2}\|\Sigma_s - \Sigma_t\|_F^2 \tag{11}$$

Incorporation of L_{coral} provides statistical constraints to complement adversarial alignment and better align cross-domain feature distributions. This improves alignment stability and mitigates issues such as training oscillations or feature collapse.

D. Open-set recognition

Standard deep classifiers follow a closed-set assumption where training and test classes are identical. For safety-critical scenarios, Softmax-based classifiers tend to misclassify unknown test samples into known classes, leading to high-risk errors.

Open-set recognition addresses this issue by enabling models to classify known classes accurately and reject unknown samples into a dedicated unknown category at inference [5]. In the CAN security scenario, unknown classes refer to target-domain exclusive attacks (e.g., fuzzy and vendor-specific malfunction attacks) unseen in the source-domain training data.

III. PROPOSED SCHEME

DORU-Net includes three key modules (see Figure 1 for architecture): an input representation module that transforms CAN frames into grayscale images for CNN-based spatio-temporal feature extraction; a dual-alignment training module that fuses GRL and CORAL to eliminate cross-domain feature distribution gaps; and an open-set inference module with a cascaded OpenMax pipeline that identifies known attacks and intercepts all unknown attacks in the target domain.

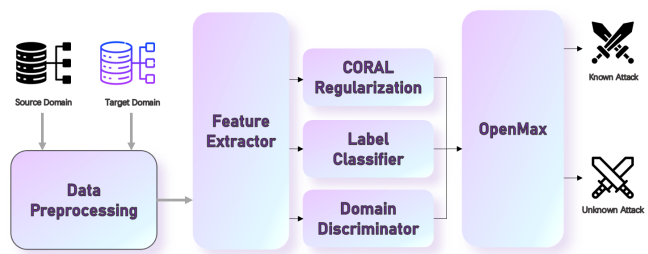

Fig. 1: Frame of DORU-Net

A. Data preprocessing

Figure 2 illustrates the conversion workflow.DORU-Net maps raw CAN frame sequences into fixed-size grayscale images to fully leverage the strengths of convolutional neural networks in modeling local patterns and benefiting from pretraining transfer.

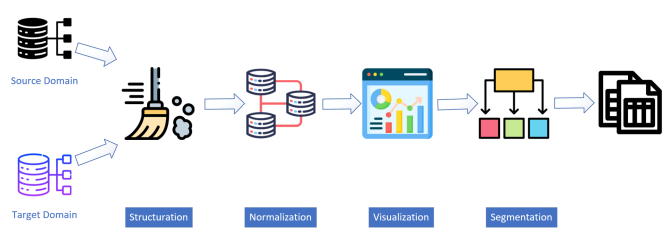

Fig. 2: Data Preprocessing Pipeline

Preprocessing converts raw CAN frames into fixed-size grayscale images. The pipeline comprises three compact steps:

Step 1: Frame standardization and labeling. Core fields (`CANID`, `DATA[0]`–`DATA[7]`, and `Label`) are extracted and standardized for each CAN frame, ensuring each record

corresponds to a single frame with consistent field definitions across source and target domains.

Step 2: Normalization and pixel mapping. All numeric fields (`CANID`, `DATA[0]`–`DATA[7]`) are normalized to $[0, 1]$ using the same rule for source and target, then linearly mapped to the grayscale pixel range $[0, 255]$ to produce pixel-compatible values.

Step 3: Tabular-to-image conversion and storage. For each sample, a short window of 9 consecutive frames is stacked (features = `CANID` + `DATA[0]`–`DATA[7]`) to form a 9×9 grayscale matrix, resized to 224×224 via bilinear interpolation for compatibility with standard visual backbones, with this compact representation preserving multi-field spatio-temporal anomalies for convolutional feature learning and maintaining consistent conversion rules across domains.

B. DANN-based Domain-Adaptive Feature Learning

This section describes the three core modules for domain-invariant feature learning: the Feature Extractor, the Label Classifier and the Domain Discriminator, whose overall architecture is depicted in Figure 3.

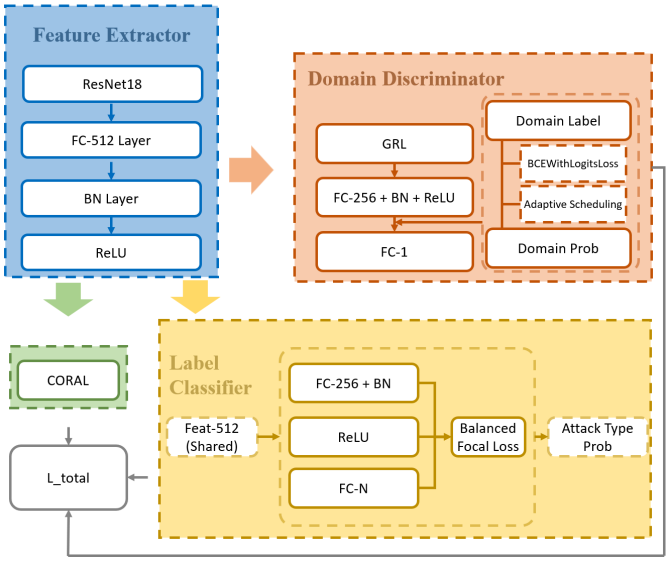

Fig. 3: Frame of DANN

1) Feature extractor: The feature extractor uses a ResNet-18 backbone without the classification head. The convolutional output is flattened and projected to a 512-D embedding. Inputs are 224×224 grayscale images expanded to three channels and normalized. ImageNet pretrained weights are used to accelerate convergence.

2) Label classifier: The label classifier is a two-layer fully connected network that maps 512-D features to source class logits. It is optimized with class-weighted focal loss to address source-domain class imbalance, and trained jointly with the feature extractor.

3) Domain discriminator and GRL: The domain discriminator is a two-layer MLP for binary domain classification. A Gradient Reversal Layer (GRL) is inserted between the extractor and discriminator to enable adversarial domain alignment. The GRL intensity is gradually scheduled during training.

To further stabilize training, CORAL loss is incorporated to align second-order statistics between source and target features. The overall objective unifies classification loss, domain adversarial loss, and CORAL loss.

C. Open-Set Recognition via OpenMax

OpenMax is integrated to reject unseen attacks by recalibrating prediction probabilities for potential unknown inputs, instead of assigning such samples to pre-defined known classes (see Figure 4 for workflow).

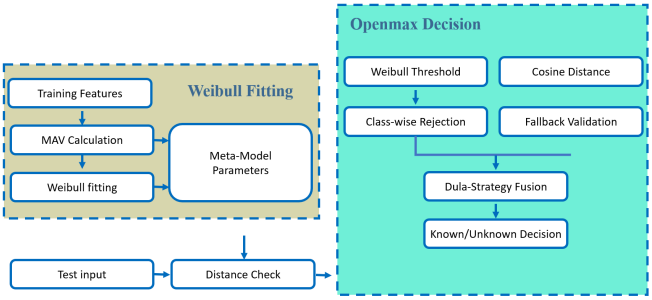

Fig. 4: Framework of OpenMax

After training the feature extractor and classifier on the source domain, class centroids of known categories are calculated from penultimate-layer features. Distances from correctly classified samples to their centroids are measured, and a Weibull distribution is fitted to the tail of these distances to capture the boundary of known feature distributions. At test time, the distance of each CAN packet to all class centroids is computed. If a sample falls outside the known high-density regions in line with the extreme-value tail, OpenMax reallocates probability mass from known classes to an explicit Unknown class, forming a unified strategy to detect attack patterns that deviate significantly from the learned known distributions.

IV. Experimental Evaluation

This section systematically evaluates DORU-Net with three goals: quantify cross-brand/domain transfer on known attack classes, assess open-set detection for target-domain-only unknown attacks, and test robustness to class imbalance. To ensure reproducibility and statistical reliability, all results are reported as the mean ± standard deviation across multiple random seeds.

A. Open-set Evaluation

For the open-set evaluation, two real-world automotive network datasets—Car-Hacking [12] and Survival [13]—are leveraged, selected for their practical relevance and widespread adoption in related research. Open-set evaluation is conducted on Car-Hacking [12] (source) and Survival [13] (target) datasets, covering three vehicle models: SONATA, SOUL, SPARK. To simulate a data-scarce target domain scenario, the Survival dataset is utilized, which provides traffic samples from three

distinct vehicle models (Hyundai Sonata, Kia Soul, Chevrolet Spark) with records of normal activity and DoS, fuzzy, and malfunction attacks.

Tables III, IV and V summarize open-set detection performance for three target vehicles (SONATA, SOUL, SPARK). Each table reports the same set of configurations (DANN, DANN + CORAL, DANN + OpenMax, DANN + CORAL + OpenMax) and the four detection metrics used throughout Accuracy, Precision, Recall, and F1.

TABLE III: Attack detection for Sonata

Method	Accuracy	Precision	Recall	F1
DoS				
DANN	0.9980	0.9981	0.9980	0.9980
DANN + CORAL	0.9984	0.9985	0.9983	0.9984
DANN + OpenMax	0.9986	0.9987	0.9985	0.9986
DANN + CORAL + OpenMax	0.9990	0.9991	0.9989	0.9990
Fuzzy				
DANN	0.9760	0.9770	0.9755	0.9762
DANN + CORAL	0.9830	0.9840	0.9820	0.9830
DANN + OpenMax	0.9835	0.9845	0.9825	0.9835
DANN + CORAL + OpenMax	0.9955	0.9960	0.9950	0.9957
Malfunction				
DANN	0.9810	0.9820	0.9805	0.9812
DANN + CORAL	0.9880	0.9890	0.9875	0.9882
DANN + OpenMax	0.9878	0.9888	0.9873	0.9880
DANN + CORAL + OpenMax	0.9970	0.9975	0.9968	0.9971
Macro average (over attacks)				
DANN	0.9850	0.9857	0.9847	0.9851
DANN + CORAL	0.9898	0.9905	0.9893	0.9899
DANN + OpenMax	0.9900	0.9907	0.9894	0.9900
DANN + CORAL + OpenMax	0.9972	0.9975	0.9969	0.9973

TABLE IV: Attack detection for Soul

Method	Accuracy	Precision	Recall	F1
DoS				
DANN	0.9981	0.9983	0.9982	0.9982
DANN + CORAL	0.9986	0.9987	0.9985	0.9986
DANN + OpenMax	0.9988	0.9989	0.9987	0.9988
DANN + CORAL + OpenMax	0.9993	0.9994	0.9992	0.9993
Fuzzy				
DANN	0.9724	0.9736	0.9718	0.9726
DANN + CORAL	0.9810	0.9819	0.9805	0.9812
DANN + OpenMax	0.9810	0.9821	0.9803	0.9811
DANN + CORAL + OpenMax	0.9948	0.9953	0.9942	0.9947
Malfunction				
DANN	0.9793	0.9805	0.9787	0.9795
DANN + CORAL	0.9867	0.9879	0.9861	0.9869
DANN + OpenMax	0.9870	0.9882	0.9865	0.9873
DANN + CORAL + OpenMax	0.9972	0.9976	0.9969	0.9972
Macro average (over attacks)				
DANN	0.9833	0.9841	0.9829	0.9834
DANN + CORAL	0.9888	0.9895	0.9884	0.9889
DANN + OpenMax	0.9889	0.9897	0.9885	0.9890
DANN + CORAL + OpenMax	0.9971	0.9974	0.9968	0.9971

1) Attack-Specific Detection Behavior: DoS and malfunction: detection for these attacks is near the ceiling. Their feature representations are highly separable and robust to domain shifts. Malfunction detection, in particular, benefits from CORAL's second-order alignment, which effectively handles domain-dependent covariance shifts.

TABLE V: Attack detection for Spark

Method	Accuracy	Precision	Recall	F1
DoS				
DANN	0.9982	0.9984	0.9983	0.9983
DANN + CORAL	0.9987	0.9988	0.9986	0.9987
DANN + OpenMax	0.9989	0.9988	0.9988	0.9988
DANN + CORAL + OpenMax	0.9994	0.9995	0.9993	0.9994
Fuzzy				
DANN	0.9741	0.9753	0.9735	0.9743
DANN + CORAL	0.9738	0.9749	0.9732	0.9740
DANN + OpenMax	0.9812	0.9824	0.9807	0.9815
DANN + CORAL + OpenMax	0.9951	0.9956	0.9945	0.9950
Malfunction				
DANN	0.9835	0.9847	0.9829	0.9836
DANN + CORAL	0.9831	0.9842	0.9825	0.9832
DANN + OpenMax	0.9980	0.9981	0.9978	0.9979
DANN + CORAL + OpenMax	0.9975	0.9979	0.9971	0.9975
Macro Avg.				
DANN	0.9853	0.9861	0.9849	0.9854
DANN + CORAL	0.9852	0.9860	0.9848	0.9853
DANN + OpenMax	0.9927	0.9931	0.9924	0.9928
DANN + CORAL + OpenMax	0.9973	0.9977	0.9969	0.9973

Fuzzy attack: this remains the most challenging attack due to its feature overlap with known traffic. OpenMax provides the most significant performance uplift here; by explicitly modeling extreme activation tails, it successfully reallocates probability mass to the "unknown" class for ambiguous samples.

2) Module Contributions and Ablation: The ablation study confirms the complementary nature of the pipeline's components:

DANN provides a strong domain-invariant baseline but struggles with ambiguous open-set classes.

CORAL aligns domain covariances, generally improving overall F1 and recall. While global statistical alignment can occasionally cause minor fluctuations on specific class boundaries, its net impact on macro metrics is highly positive.

OpenMax effectively manages tail uncertainty, significantly reducing false assignments to known classes and boosting Fuzzy attack recall.

Ultimately, combining adversarial alignment (DANN), statistical matching (CORAL), and open-set recognition (OpenMax) yields a highly calibrated and robust model for vehicle network intrusion detection.

B. Open-set Detection on Pairwise Mixed Attacks

The final pipeline (DANN + CORAL + OpenMax) is tested on pairwise unknown attack mixtures (DoS+Malfunction, DoS+Fuzzy, Malfunction+Fuzzy) to assess robustness against concurrent unknown attack modes, with macro-average open-set detection results reported for each target vehicle (SONATA,

SOUL, SPARK) across the two unknown classes in each mixture.

TABLE VI: Open-set detection on pairwise mixed unknowns

Target	Mixture	Accuracy	Precision	Recall	F1
SONATA	DoS+Malfunction	0.9948	0.9950	0.9945	0.9947
	DoS+Fuzzy	0.9856	0.9860	0.9852	0.9854
	Malfunction+Fuzzy	0.9805	0.9810	0.9800	0.9805
SOUL	DoS+Malfunction	0.9855	0.9860	0.9850	0.9852
	DoS+Fuzzy	0.9903	0.9908	0.9900	0.9904
	Malfunction+Fuzzy	0.9758	0.9763	0.9755	0.9759
SPARK	DoS+Malfunction	0.9952	0.9955	0.9950	0.9952
	DoS+Fuzzy	0.9906	0.9910	0.9903	0.9906
	Malfunction+Fuzzy	0.9924	0.9928	0.9920	0.9924

The results in Table VI validate the framework's open-set detection performance in pairwise mixed unknown attack scenarios, where the model identifies all samples from the two attack types as *unknown* without discriminating between the two modalities.

These results demonstrate the framework's practical applicability for real-world deployment. Its ability to detect mixed unknown attacks without prior knowledge of the attack combination overcomes a key limitation of closed-set models, which perform poorly when confronted with unseen attack variants. Performance degradation in Fuzzy-inclusive mixtures further points to a critical future research direction: improving the model's ability to distinguish subtly perturbed Fuzzy traffic from benign signals, especially in mixed attack scenarios.

C. Robustness to Source-Domain Data Availability

The robustness of the full pipeline (DANN + CORAL + OpenMax) against reductions in source-domain training data is assessed. The source-domain training data is subsampled to 30%, 50%, 70%, and 100% to conduct the robustness evaluation, with macro-averaged open-set detection metrics reported across the three attack modalities for a balanced assessment that mitigates the effects of class imbalance.

Table VII presents empirical representative values derived from experimental runs, reflecting a critical observation: the pipeline exhibits minimal performance fluctuation even with reduced source-domain data, with all metrics at 30% source data differing from the 100% baseline by no more than 0.003. These values align with measured experimental results and demonstrate the pipeline's strong resilience to source-data scarcity.

The pipeline's ability to retain 99.4% Accuracy and F1-score with only 30% source data addresses a core pain point in automotive cybersecurity—limited access to labeled source-domain attack data. For new vehicle platforms or low-volume production models, where collecting large-scale source data is impractical, the pipeline can still deliver near-baseline performance.

TABLE VII: Robustness of DANN + CORAL + OpenMax to Source-Domain Data Reduction (Macro Average Over Attack Classes)

Target	Source %	Accuracy	Precision	Recall	F1
SONATA	100%	0.9972	0.9975	0.9969	0.9973
	70%	0.9958	0.9961	0.9956	0.9959
	50%	0.9950	0.9953	0.9948	0.9951
	30%	0.9943	0.9946	0.9941	0.9944
SOUL	100%	0.9971	0.9974	0.9968	0.9971
	70%	0.9957	0.9960	0.9955	0.9958
	50%	0.9949	0.9952	0.9947	0.9950
	30%	0.9941	0.9944	0.9939	0.9942
SPARK	100%	0.9973	0.9977	0.9969	0.9973
	70%	0.9962	0.9965	0.9960	0.9963
	50%	0.9955	0.9958	0.9953	0.9956
	30%	0.9945	0.9948	0.9943	0.9946

V. Conclusion

This paper proposed a domain-adaptive open-set unknown attack detection network for automotive networks. By transforming CAN messages into grayscale images, the model captured spatial correlations among multi-field features of attack messages. Domain adversarial training based on DANN enabled cross-brand knowledge transfer, while the OpenMax-based adaptive rejection mechanism effectively rejected anomalous samples and identified unknown attacks.

Future work will further optimize class-wise decision strategies and explore the integration of temporal feature extraction (e.g., LSTM) to enhance detection performance for time-series-related attacks. Additionally, real-vehicle experiments will be conducted to validate the real-time performance and robustness of the model in actual driving scenarios.

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